

Prediction figures and validation results

Computed from **artifacts_final/model.pt** using the local MLII MATLAB dataset. This package provides seven figures, exact result tables, and traceable validation predictions for manuscript preparation.

Measure	Result
Correct / validation fragments	174 / 200
Accuracy	87.00%
Weighted precision	86.40%
Weighted recall	87.00%
Weighted F1	86.22%
Macro F1	76.09%
Balanced accuracy	76.79%
Macro one-versus-rest ROC AUC	0.9907
Macro average precision	0.9151
Trainable parameters	27,633

Interpretation to retain in the paper

These are **internal, fragment-level validation results with data leakage**. All 200 validation fragments have a filename-derived recording ID also present in training (44 shared recording IDs); three validation waveforms also exactly match training waveforms. The IDs are recording proxies, not verified patient identities. These scores do not establish performance on unseen recordings or patients.

The evaluated system is the implemented 1-D residual-attention baseline. The full proposed neuro-cardiac pipeline, treatment recommendation, and full ResilienceNet framework are not evaluated here. Figures include both correct predictions and failure cases.

TABLE 1 / CLASSIFICATION

Performance by class

Class scores are percentages. Support n is the number of validation fragments. Specificity and continuous-score ranking metrics are supplied in the accompanying CSV table.

Class	n	Precision	Recall	F1
1 NSR	57	91.07	89.47	90.27
2 APB	13	55.56	76.92	64.52
3 AFL	4	80.00	100.00	88.89
4 AFIB	27	96.43	100.00	98.18
5 SVTA	3	0.00	0.00	0.00
6 WPW	4	100.00	100.00	100.00
7 PVC	26	79.17	73.08	76.00
8 Bigeminy	11	71.43	90.91	80.00
9 Trigeminy	3	100.00	33.33	50.00
10 VT	2	100.00	100.00	100.00
11 IVR	2	50.00	50.00	50.00
12 VFL	2	100.00	100.00	100.00
13 Fusion	2	0.00	0.00	0.00
14 LBBBB	21	100.00	100.00	100.00
15 RBBBB	12	100.00	91.67	95.65
16 SDHB	2	100.00	100.00	100.00
17 PR	9	100.00	100.00	100.00

Fusion and SVTA each have zero recall. Rare categories often have only two to four validation fragments, so seemingly perfect class scores are based on very small counts. The difference between weighted F1 (86.22%) and macro F1 (76.09%) shows the influence of class imbalance.

ROC AUC measures ranking and may remain high despite argmax misclassification. For example, Trigeminy has perfect one-versus-rest AUC and average precision on this split, while only one of its three examples is correctly classified.

Confusion matrix

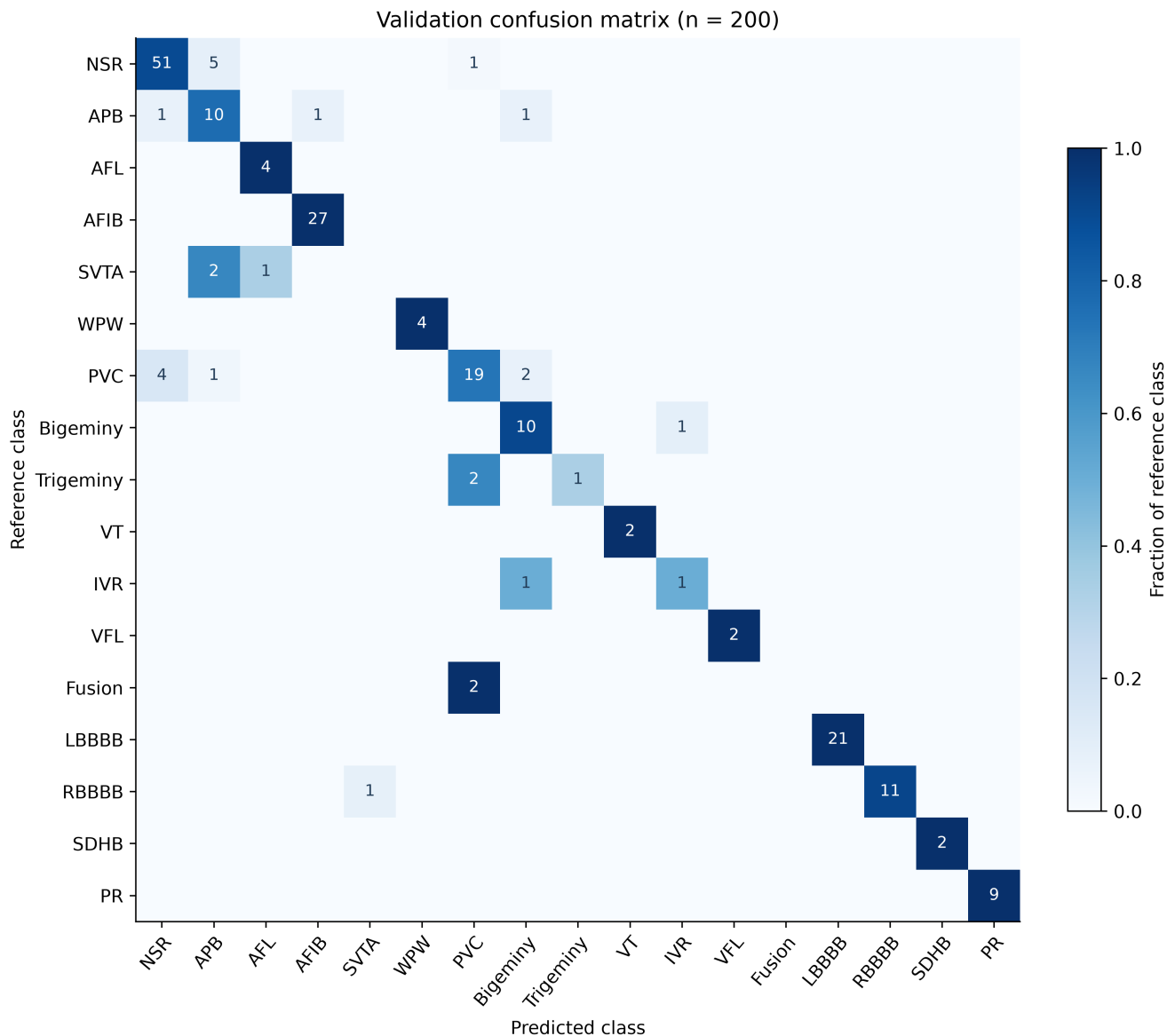


Figure 1. Confusion matrix for 200 validation ECG fragments. Cell annotations are recording-fragment counts; color is normalized within each reference class. The diagonal contains 174 correct predictions. No independent patient or recording holdout was used.

Standalone files: fig01_confusion_matrix.png (400 dpi) and .svg (editable vector).

Per-class precision, recall and F1

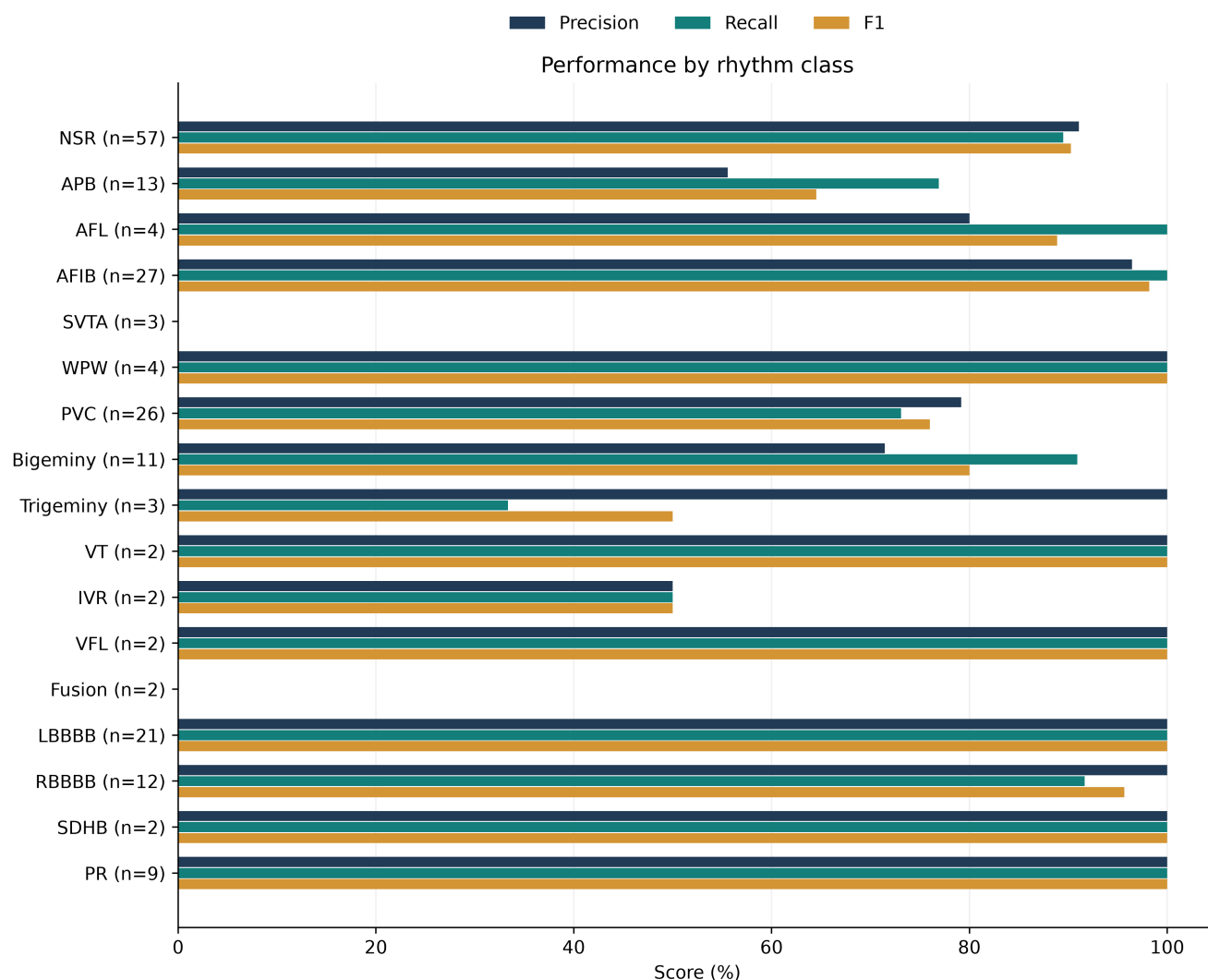


Figure 2. One-versus-rest precision, recall and F1 scores on the validation split. Parentheses show validation support. Fusion and SVTA have zero recall; several categories have only two to four examples. Undefined precision is assigned zero. Macro F1 gives equal class weight; weighted F1 weights each class by validation support.

Standalone files: fig02_per_class_scores.png (400 dpi) and .svg (editable vector).

ROC and precision-recall curves

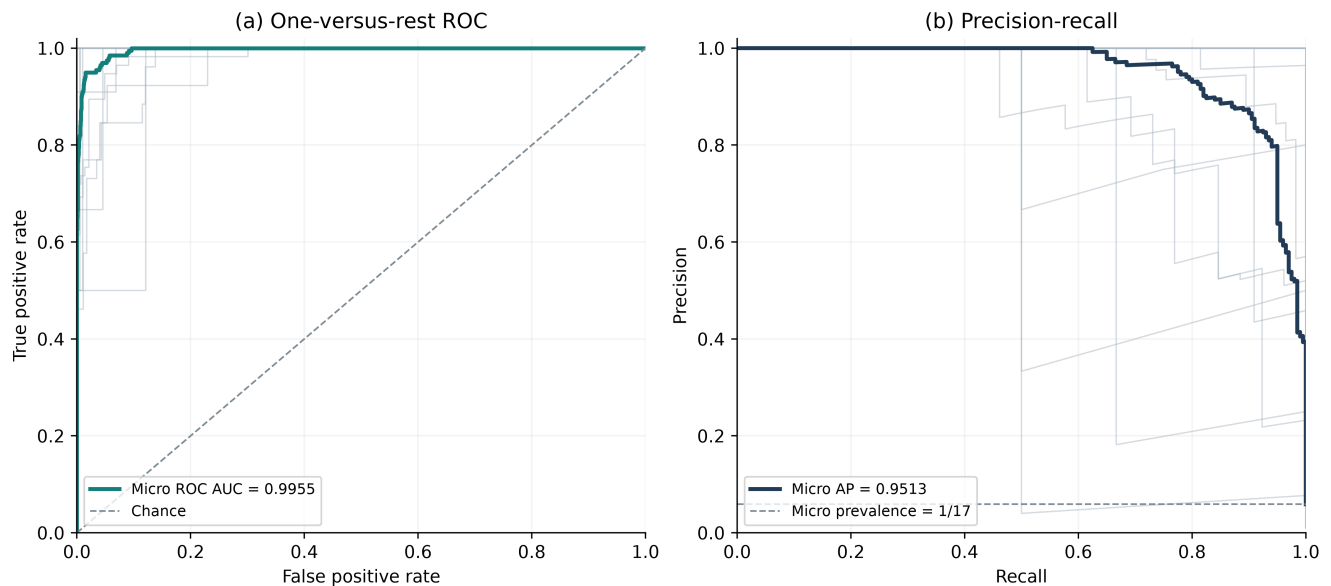


Figure 3. One-versus-rest ROC and precision-recall curves from continuous softmax scores. Pale lines show individual classes; bold lines show micro-averaged curves. Mean class ROC AUC is 0.9907, and mean class average precision (AP) is 0.9151. AP is computed as a recall-weighted sum of precision, not trapezoidal PR area. Ranking metrics do not replace argmax classification scores, particularly for small classes.

Standalone files: fig03_roc_precision_recall.png (400 dpi) and .svg (editable vector).

Correct ECG prediction examples

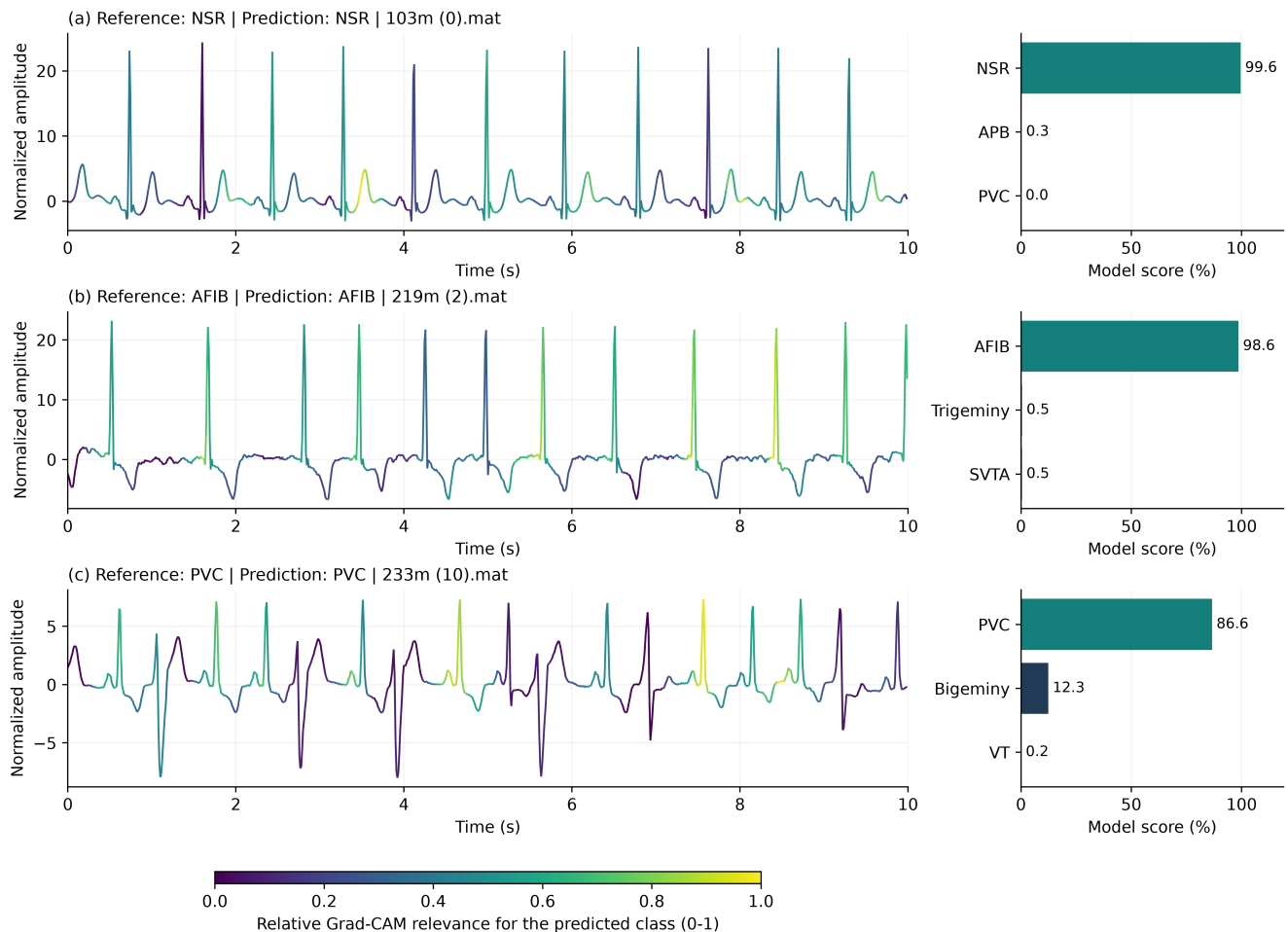


Figure 4. One correct NSR, AFIB and PVC example is selected at the median predicted-class score among correct examples of that class (upper median when even). All examples are from validation. Waveforms are filtered, normalized, and sampled at 90 Hz for the model. Color denotes within-example Grad-CAM relevance, not a clinical annotation; each map is independently normalized. Right panels show the three largest uncalibrated softmax scores. Teal bars denote the reference class when it occurs among the top three.

Standalone files: fig04_correct_predictions.png (400 dpi) and .svg (editable vector).

High-confidence prediction errors

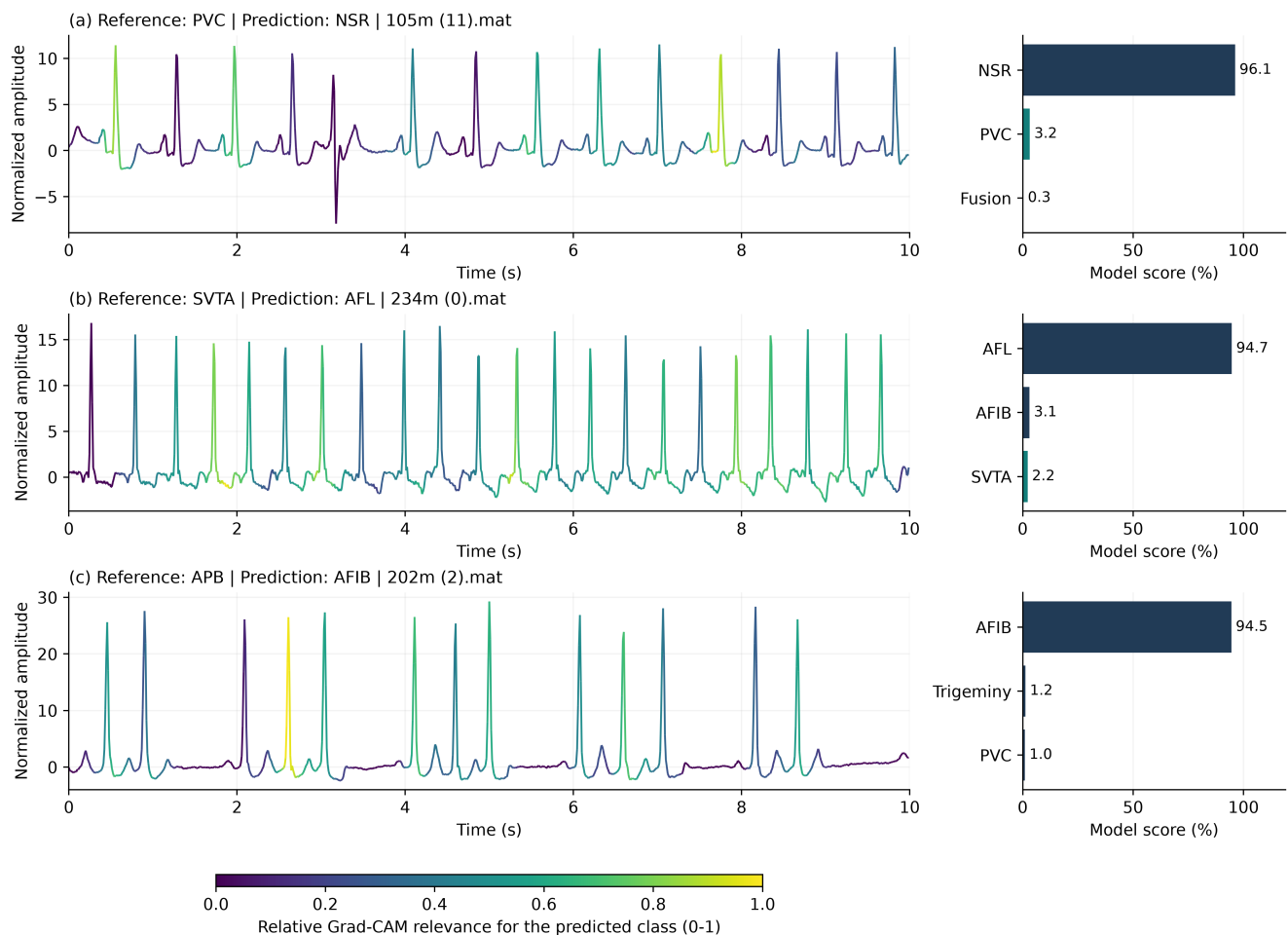


Figure 5. The three highest-score misclassifications with distinct reference classes are shown in descending predicted-score order to illustrate failure cases; this is a deliberate error-focused selection. All examples are from validation. Waveforms are filtered, normalized, and sampled at 90 Hz for the model. Color denotes within-example Grad-CAM relevance, not a clinical annotation; each map is independently normalized. Right panels show the three largest uncalibrated softmax scores. Teal bars denote the reference class when it occurs among the top three.

Standalone files: fig05_prediction_errors.png (400 dpi) and .svg (editable vector).

FGSM robustness curve

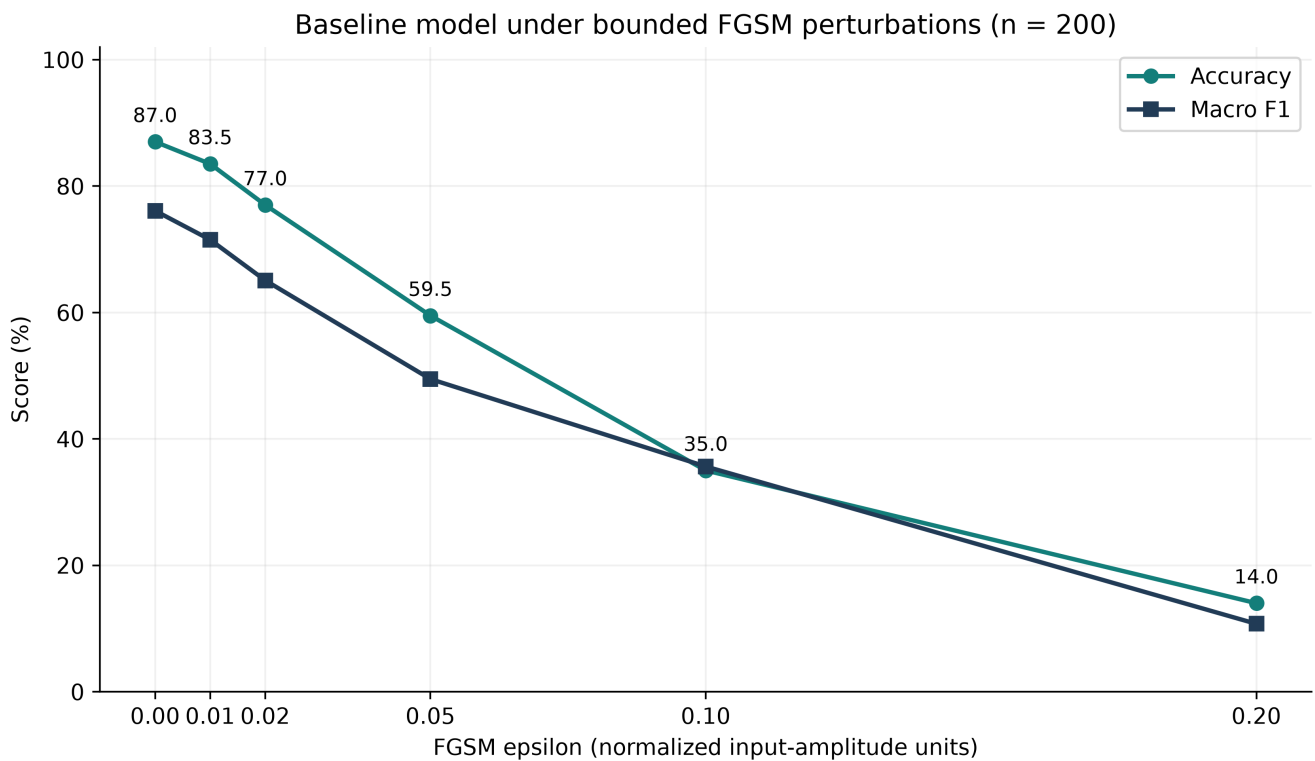


Figure 6. White-box, untargeted FGSM evaluation on the same 200 validation fragments. Perturbations use the gradient of cross-entropy with respect to the normalized 90 Hz model input, with no arbitrary amplitude clipping. Epsilon is an input-amplitude bound. The evaluated checkpoint is the original baseline, not an adversarially trained or complete ResilienceNet model. These scores measure only this specific attack and are not a security certificate.

Standalone files: fig06_adversarial_robustness.png (400 dpi) and .svg (editable vector).

Dataset and validation support

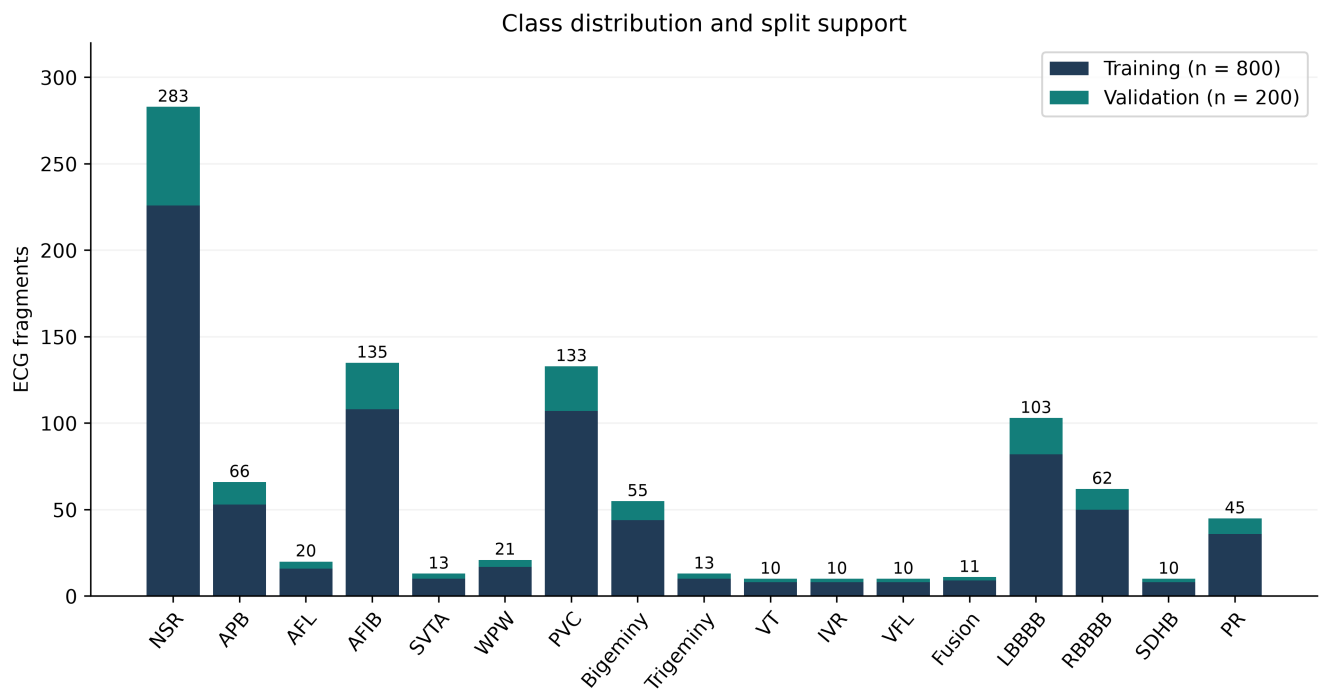


Figure 7. Distribution of 1,000 source ECG fragments across 17 classes, with stacked counts for the reconstructed stratified training and validation partitions. Values above bars are total class counts. The validation sample sizes range from 2 to 57 fragments, limiting precision of class-specific estimates.

Standalone files: fig07_class_distribution.png (400 dpi) and .svg (editable vector).

Evaluation details

Data and preprocessing

The source provides 1,000 labelled 10-second MII ECG fragments sampled at 360 Hz. This implementation applies fourth-order 0.5-45 Hz Butterworth filtering using zero-phase SOS filtering, median subtraction and scaled median absolute deviation normalization. Every fourth sample is retained to match the existing model's 90 Hz input. This subsampling design is retained for checkpoint compatibility; changing it requires retraining.

Split and checkpoint

A stratified 80/20 fragment split is reconstructed with scikit-learn `train_test_split` and `random_state=42` in the same sorted file order used by the original loader. The legacy checkpoint does not store split metadata, so the seed is inferred from the original default and confirmed by exact reproduction of saved accuracy and weighted metrics. No retraining was performed for these results.

Predictions and explanation

Classification uses the maximum softmax score. Precision, recall and F1 are computed per class and summarized with macro and support-weighted averages. Undefined precision is assigned zero. One-versus-rest ROC AUC and average precision use continuous scores. Grad-CAM weights the final residual-block activations by the temporal average of the predicted-class gradient, applies ReLU, interpolates to the model-input grid, and scales each map to 0-1. These maps have not been clinically validated.

Adversarial test

FGSM uses epsilon times the sign of the cross-entropy input gradient. The normalized inputs have no fixed amplitude range, so no extra clipping is applied. The entire sweep uses the same baseline checkpoint and validation set. At `epsilon=0.05`, accuracy is 59.5%, versus 87% for clean inputs. The earlier 47% estimate used extra clipping and is superseded. This test does not evaluate encryption security or certify robustness.

Files and reproducibility

`results.json` records software versions and checkpoint SHA-256; `split_manifest.csv` records file hashes, indices and split membership. `validation_predictions.csv` includes every reference label, prediction and class score. `cross_split_duplicates.csv` identifies the three exactly duplicated waveform pairs. `example_selection.csv` specifies the figure selection rules. Learning curves are omitted because no per-epoch validation history was saved.

References

Plawiak, P. (2017). ECG signals (1000 fragments), Mendeley Data V3, CC BY 4.0. doi:10.17632/7dybx7wyfn.3.

Selvaraju et al. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. arXiv:1610.02391.

Goodfellow, Shlens and Szegedy. Explaining and Harnessing Adversarial Examples. arXiv:1412.6572.